



Genes involved in regeneration of worms

Harvard study discovered noncoding DNA that controls activation of a "master control gene" that is called early growth response in worms. <https://digit.in/apr19-13>



Cut-off that low-Aspirin dose now

New guidelines show risk of bleeding more than heart benefits among healthy adults on low-dose aspirin. <https://digit.in/apr19-14>

SPACE ROAR

The ancient universe is screaming, and scientists don't know why

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In July 2006, a team of researchers from a NASA facility in Texas sent up a balloon loaded with instruments to study stars born 13 billion years ago, at the dawn of the universe. The experiment on board was called Absolute Radiometer for Cosmology, Astrophysics, and Diffuse Emission or ARCADE. The purpose of the instrument was to study radio waves, which fall beyond the infrared and microwave sections of the electromagnetic spectrum. They have low energies, low frequencies, and are extremely elongated.

Studying the objects in the skies through their radio emissions is called radio astronomy. Back in 1928, a physicist by the name of Karl Jansky joined Bell Laboratories. At the facility, he built a radio antenna mounted on a turntable. Turning the antenna allowed Jansky to pinpoint the source of radio waves, and the instrument earned the name of "Jansky's merry-go-round". Jansky tracked three types of radio static on the instrument, one associated with nearby thunderstorms, another associated with

distant thunderstorms, and the third of unknown origin. There was an unexplained, seemingly celestial source of radio hiss. Jansky spent a year tracking the signal, and it had a periodicity roughly following a day, rising to a peak maximum and falling off. The signal repeated every 23 hours and 56 minutes, which was the characteristic of the stars. Eventually, Jansky realised that the peak maximum coincided with the center of the Milky Way galaxy, in the direction of the constellation of Sagittarius. It was the first time that radio waves had been observed from a celestial body, and the discovery gave rise to the field of radio astronomy. Over the years, stars and galaxies have been identified as sources of radio waves. There are also some exotic celestial objects that are particularly active in the radio frequencies. These include galaxies with bright centers called active galactic nuclei, such as radio galaxies, quasars and blazars. Highly magnetized and rapidly spin-

ning neutron stars called pulsars are also luminous or loud in the radio wavelengths. In the visible spectrum, the interstellar space appears perfectly dark. However, in the radio wavelengths, there is a uniform glow in the universe. This is known as the cosmic microwave background (CMB), which is uniform in all directions, and is the oldest electromagnetic radiation measured in the universe, from the time when the first atoms were being formed. The discovery of the CMB was a triumph of radio astronomy, and provided evidence for the Big Bang theory of the origin of the universe.

THE UNEXPECTED NOISE

The radiation from the earliest stars only reach Earth as radio waves, because of how incredibly distant they are, as well as the expansion of the universe. Radio astronomy is particularly suitable for studying the earliest time

